Company Growth and Workforce Analytics Using Python

Project Documentation Report

Dataset: Inc 5000 Companies dataset
Domain: Finance / Business Analytics
Dataset size recorded in notebook: 502 rows × 14 original columns

Prepared from the uploaded Jupyter Notebook and its recorded outputs.

1. Project Overview

This project analyzes company growth and workforce information from the Inc 5000 Companies dataset recorded in the notebook. Using Python, the notebook inspects the dataset, checks data quality, transforms revenue into a numeric format, calculates workforce changes, and creates additional features for company size and company age. Exploratory data analysis (EDA) is then used to summarize company and industry distributions and review numeric measures.

The notebook's focus is on understanding company growth, workforce expansion, industry representation, and geographical distribution. The report describes the operations and results that are visible in the uploaded notebook; it does not assume that every proposed analysis question has already been answered.

2. Data Source

Dataset name: Inc 5000 Companies dataset
Source stated in notebook: Maven Analytics Data Playground
Source link: https://mavenanalytics.io/data-playground
Dataset description: Company information for entries on the 2019 Inc. 5000 list.
Original dimensions: 502 rows and 14 columns.
Geographical coverage stated in notebook: 43 states, 293 cities, and 54 distinct metropolitan-area labels.
Year range stated in notebook: founding years from 1909 to 2015.

The notebook loads a CSV file from a GitHub raw-content URL. Maven Analytics is recorded as the original dataset source, while the notebook's code uses the GitHub-hosted CSV for loading.

3. Problem Statement

Company growth and workforce information contains multiple fields such as revenue, growth percentage, industry, employee counts, founding year, and location. Without systematic preparation and analysis, it is difficult to compare companies, understand workforce changes, or summarize the industries and places represented in the dataset. This project uses Python-based data cleaning, feature engineering, and exploratory analysis to organize the data and make these comparisons easier to investigate.

4. Project Objectives

1. Analyze company growth: Identify which company has the highest revenue growth percentage among the top 10 companies.

2. Compare industry performance: Investigate which industry generates the highest total revenue.

3. Evaluate workforce growth: Identify the company with the highest employee growth relative to its previous workforce.

4. Analyze geographical distribution: Find the state or city with the largest number of companies.

5. Identify high-growth business trends: Investigate whether revenue and growth percentage appear to be related.

6. Analyze revenue per employee: Compare companies using revenue relative to employee count as a workforce-productivity indicator.

7. Study company age and growth: Examine company counts by founding year and compare growth percentages of newer and older companies.

8. Identify workforce expansion leaders: Investigate which industries have the largest workforce.

9. Analyze business growth by location: Explore whether company location is associated with revenue or growth performance.

10. Compare revenue and workforce size: Determine whether larger workforces tend to occur alongside higher revenue.

> These are the ten objectives documented in the notebook. The recorded code shown in the notebook performs initial EDA and data preparation; not every objective has a displayed final result yet.

5. Attribute (Column / Feature Details)

The notebook starts with 14 original fields. It renames five columns and adds five engineered columns, producing 19 columns in the final dataframe.

Column / feature	Description / role
Rank	Company rank in the source list.
Profile	Company profile link.
Name → Company_name	Company name; renamed for clarity.
url	Company URL field provided by the source.
state	State associated with the company.
Revenue	Revenue represented as text, for example “36.9 Million”.
Growth_% → Growth_Percentage	Company growth percentage from the source.
Industry	Industry classification of the company.
workers → Current_Workers	Current employee/workforce count in the dataset.
previous_workers → Previous_Workers	Previous workforce count.
Founded	Year the company was founded.
yrs_on_list → Years_On_List	Number of years on the list.
Metro	Metropolitan-area label; missing/“Unknown” values are replaced using city.

Column / feature	Description / role
city	City associated with the company.
Revenue_Million	Engineered numeric revenue value in millions; original Revenue column is retained.
Employee_Change	Current_Workers minus Previous_Workers.
Employee_Growth_Percentage	Employee_Change divided by Previous_Workers, multiplied by 100; rounded to 2 decimal places.
Company_Size	Category based on current workers: Small (<50), Medium (50–249), Large (≥250).
Company_Age_2015	Reference-year age calculated as 2015 minus Founded.

6. Tools & Technologies

Technology	Usage
Python	Main programming language for data preparation and analysis.
Jupyter Notebook	Environment used to write and execute code cells and document results.
Pandas	Loads the CSV, inspects data, handles missing values, renames columns, transforms data, and summarizes categories.
NumPy	Imported for numerical operations; the visible notebook does not show a specific NumPy operation.
Matplotlib	Creates box plots to inspect possible outliers in workforce and growth percentage.
Seaborn	Imported for visualization; no specific Seaborn chart is visible in the provided notebook cells.
CSV	Input format for the original dataset and output format for the cleaned dataset.

7. Data Preprocessing and Feature Engineering

A. Data cleaning steps

1. Load and inspect: The notebook loads the CSV with Pandas and uses head(), shape, columns, info(), dtypes, and describe() to inspect the data.

2. Check duplicates: The notebook finds 0 duplicate rows. Running drop_duplicates() leaves 502 rows.

3. Check missing values: The original dataset has 70 missing values, all in the Metro column. The notebook replaces “Unknown” with missing values and fills Metro from the city column. The final missing-value summary records 0 missing values across all 19 columns.

4. Rename columns: Name becomes Company_name; Growth_% becomes Growth_Percentage; workers becomes Current_Workers; previous_workers becomes Previous_Workers; and yrs_on_list becomes Years_On_List.

5. Inspect possible outliers: Box plots are created for Current_Workers and Growth_Percentage. These plots are used for inspection; the notebook does not show removal of outliers.

6. Convert revenue to numeric: The text “ Million” is removed from Revenue and the numeric value is converted to float in a new Revenue_Million column. The original Revenue column is kept.

B. Feature Engineering

1. Employee_Change: Calculated as Current_Workers – Previous_Workers.

2. Employee_Growth_Percentage: Calculated as (Employee_Change / Previous_Workers) × 100. The result is rounded to two decimal places. The calculation assumes Previous_Workers is non-zero.

3. Company_Size: Companies with fewer than 50 current workers are classified as Small; 50–249 as Medium; and 250 or more as Large.

4. Company_Age_2015: Calculated as 2015 – Founded. This is age relative to the reference year 2015, not the current year.

5. Save processed data: The notebook saves the processed dataframe as cleaned_company_dataset.csv.

8. Analysis and Visualisations

The uploaded notebook includes box plots for workforce and source growth percentage, along with category counts and descriptive statistics. The following figures are generated from the category counts recorded in the notebook outputs.

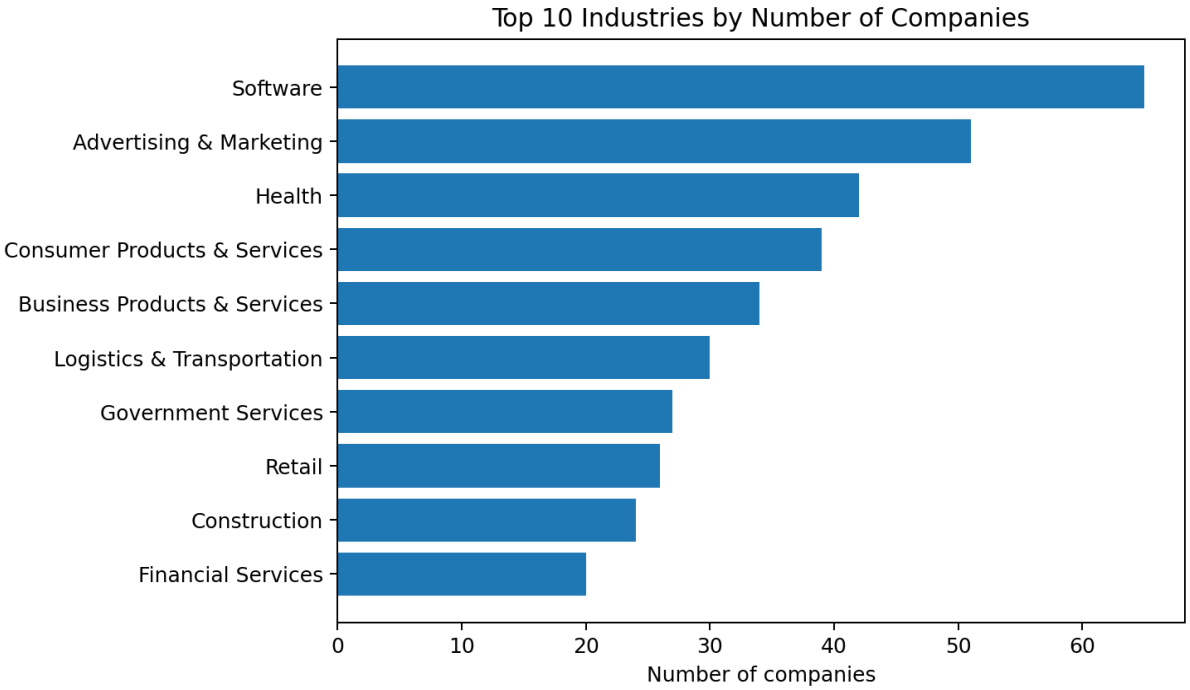


Figure 1. Top 10 industries by number of companies. Software has the largest count (65), followed by Advertising & Marketing (51) and Health (42).

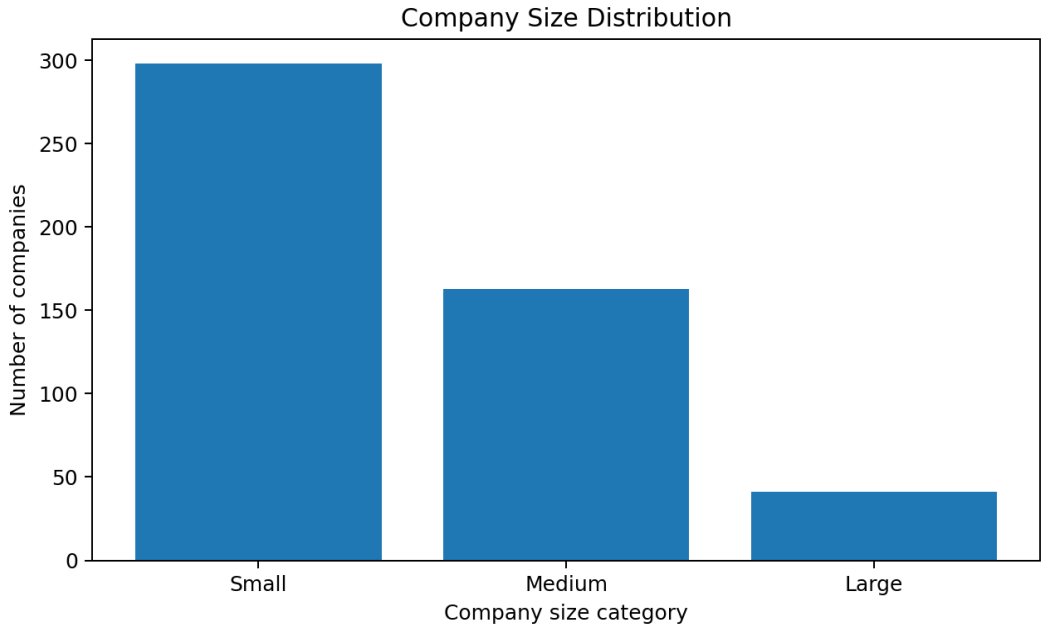


Figure 2. Company-size categories created from Current_Workers: Small 298, Medium 163, and Large 41.

Suggested next visualisations aligned with the notebook objectives: revenue by industry, top 10 company growth percentages, employee growth by company, companies by state/city, revenue versus workforce scatter plot, and growth percentage versus company age. These are recommended follow-up analyses; they are not claimed as completed results in the uploaded notebook.

9. Exploratory Data Analysis (EDA)

The notebook's initial and final EDA steps summarize the structure and numerical characteristics of the dataset.

9.1 Dataset structure

EDA item	Recorded result
Original dataset shape	502 rows × 14 columns
Final dataframe shape	502 rows × 19 columns
Original missing values	70, all in Metro
Missing values after cleaning	0 across the final dataframe
Duplicate rows	0
Industries represented	26 distinct industry labels shown in the value-count output
Company size categories	Small: 298; Medium: 163; Large: 41

9.2 Numerical summary

Measure	Mean	Median	Minimum	Maximum
Growth_Percentage	2,717.96	1,701.40	868.61	36,680.39
Current_Workers	103.56	38	0	4,572
Previous_Workers	15.71	5	1	500
Revenue_Million	24.37	8.05	2.0	726.1
Employee_Change	87.85	31	-8	4,174
Employee_Growth_Percentage	898.81	500	-100	14,700

Note: Numeric summaries are copied from the saved notebook output and rounded for readability. Revenue_Million values are in millions of dollars as interpreted by the notebook's conversion.

9.3 Main observations from the recorded outputs

• Software is the most represented industry in the dataset with 65 companies. Advertising & Marketing follows with 51, and Health with 42.
• Small companies form the largest workforce-size category: 298 of 502 companies, compared with 163 Medium and 41 Large companies.
• The median Current_Workers count is 38, lower than the mean of about 104, indicating that the workforce distribution is pulled upward by some companies with much larger workforces.
• Employee_Growth_Percentage ranges from –100% to 14,700% in the recorded summary. Very high percentages can occur when the previous workforce count is small, so the values should be interpreted with care.
• The dataset has no duplicate rows, and the final cleaning summary reports no missing values.

9.4 Limitations and interpretation

The notebook's documented objectives include several comparisons—such as total revenue by industry, revenue per employee, and relationships between revenue, growth, workforce, and location—that are not fully shown as final results in the uploaded cells. They should be calculated and visualized before drawing conclusions about those relationships. The Employee_Growth_Percentage formula also divides by Previous_Workers, so the calculation should be checked for zero denominators if the data changes.

Conclusion

The notebook establishes a reproducible starting point for company growth and workforce analysis. It loads 502 company records, cleans the Metro field, checks duplicates, converts revenue into a numeric feature, and adds employee-change, employee-growth, company-size, and company-age features. Its current EDA shows that Software is the largest represented industry and Small is the most common size category. Further analysis and visualisations are needed to answer all ten project objectives conclusively.